Intermediate Elective 2

June McNally

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Set Up

Packages

```
library(tidyverse)
library(readxl)
library(ggplot2)
library(ggtext)
library(here)
```

Data

```
aquatic_data <- read_excel("../data/aquatic_data.xlsx")
taxon_list <- read_csv("../data/taxon_list.csv")
```

Problems:

1. Explain Your Inspiration

I chose a South China Morning Post–style timeseries line chart, inspired by Steven Ponce's "Coal falls. Renewables rise." visualization, which tracks how proportions of different categories change over time on a dark background with colored lines. The Aquatic Insects sheet spans 2020–2025 with counts of invertebrate taxa across multiple sites, making a proportion timeseries ideal for showing how the relative community composition shifts year over year. Year is mapped to the x-axis, proportion of total count to the y-axis, and taxon group (Copepod, Ostracod, Cladocera, Amphipod, Chironomid) to line color.

2. Plan Your Figure

Sketch

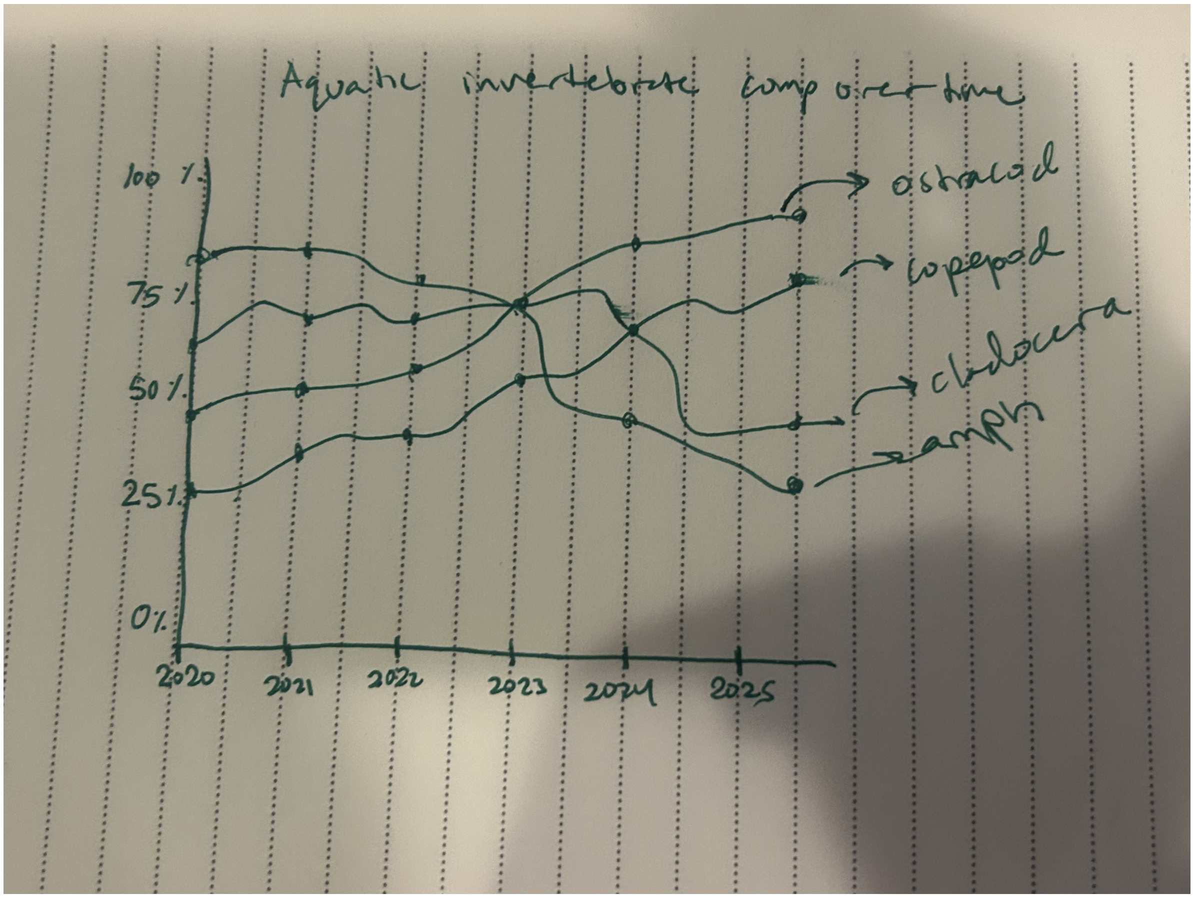

Figure 1: My figure sketch

3.Code your figure

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```r
# Load the Aquatic Insects sheet specifically - the Excel file has multiple sheets
insects <- read_excel(here("data", "aquatic_data.xlsx"),
                      sheet = "Aquatic Insects")
```

```r
# Check column names to find exact spelling of taxon columns
names(insects)
```

```
 [1] "Date data entered"
 [2] "Person Entering Data"
 [3] "Date on Vial"
 [4] "Person that sorted the sample"
 [5] "Site"
 [6] "Partial_sample_note"
 [7] "Sample Type"
 [8] "Ostracod"
 [9] "Copepod"
[10] "Hemiptera: Corixidae (boatman)"
[11] "Cladocera"
[12] "Nematode"
[13] "Diptera - TOTAL"
[14] "Dipteran sp."
[15] "Diptera: Chironomid"
[16] "Diptera: Ceratopogonidae"
[17] "Diptera : Ceratopogonidae pupa"
[18] "Diptera: Culicidae"
[19] "Diptera: Culicidae Larvae"
[20] "Diptera : Anthomyiidae"
[21] "Diptera: Ephydridae"
[22] "Diptera: Stratiomyidae"
[23] "Amphipod"
[24] "Lepidoptera-Crambidae"
[25] "Hemiptera  TOTAL"
[26] "Hemiptera- Mesoveliidae"
[27] "Hemiptera- Notonectidae"
[28] "Annelida TOTAL"
[29] "Annelida Sp."
[30] "Annelida: Oligochaete"
[31] "Annelida: Polychaete"
[32] "Ephemeroptera"
[33] "Odonate"
[34] "Coleoptera - TOTAL"
[35] "Coleoptera sp."
[36] "Coleoptera: Dytiscidae"
[37] "Coleoptera: Dytiscidae: Liodessus affinis"
[38] "Coleoptera: Dytiscidae: Agabus"
[39] "Coleoptera: Dytiscidae: Colymbetinae"
[40] "Coleoptera: Curculionidae  (Weavil)"
[41] "Coleoptera - Hydrophilidae"
[42] "Coleoptera - Tropiscernus"
[43] "Coleoperta Hydraenidae"
```

```
[44] "Coleoptera- Gyrindae- Gyrinini"
[45] "Ephemeroptera Sp."
[46] "Tipulidae Larva"
[47] "Plecoptera Sp."
[48] "Arachnida: Acari  (mite)"
[49] "Trichoptera Sp."
[50] "Collembola (Springtail)"
[51] "Gastropod - snail"
[52] "Mollusk"
[53] "Syrphidae"
[54] "Terrestrials"
[55] "Isopoda, sphaeroma"
[56] "Clam Shrimp"
[57] "Eristalis tenax larvae"
[58] "Larvae"
[59] "Crawfish"
[60] "Fish"
[61] "NZ Mudsnail"
[62] "Unknown"
[63] "Total #"
[64] "Comments"
```

```r
# Define the five key taxa we want to visualize
# Note: Chironomid is stored as "Diptera: Chironomid" in the raw data
key_taxa <- c("Copepod", "Ostracod", "Cladocera",
              "Amphipod", "Diptera: Chironomid")

# Wrangle data into long format and calculate proportion of each taxon per year
plot_data <- insects %>%
  # Keep only the date column and our five taxa of interest
  select(`Date on Vial`, all_of(key_taxa)) %>%
  # Rename for easier reference
  rename(Date = `Date on Vial`) %>%
  # Remove rows with no date (likely empty rows in the Excel file)
  filter(!is.na(Date)) %>%
  # Convert date column to Date type, then extract just the year
  mutate(Date = as.Date(Date), Year = year(Date)) %>%
  # Pivot from wide (one column per taxon) to long (one row per taxon per date)
  pivot_longer(cols = all_of(key_taxa),
               names_to = "taxon", values_to = "count") %>%
  # Replace any missing counts with 0, trim whitespace from taxon names,
  # and shorten "Diptera: Chironomid" to just "Chironomid" for cleaner labels
```

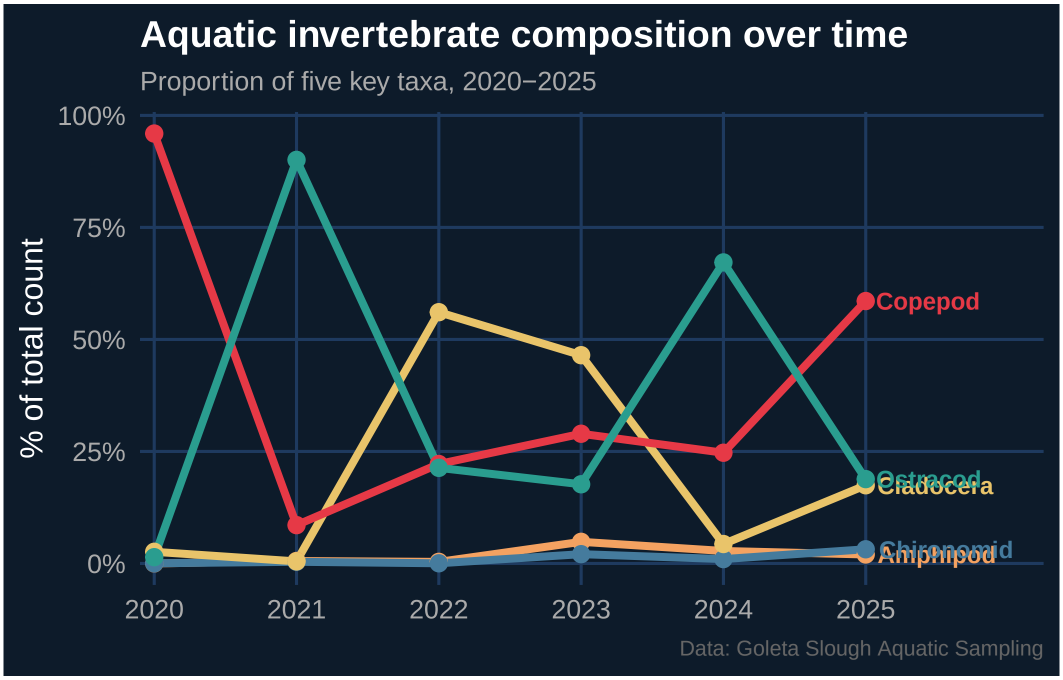
```r
  mutate(count = replace_na(count, 0),
         taxon = str_trim(taxon),
         taxon = if_else(taxon == "Diptera: Chironomid",
                         "Chironomid", taxon)) %>%
  # Sum counts for each taxon within each year across all sites
  group_by(Year, taxon) %>%
  summarise(total = sum(count, na.rm = TRUE), .groups = "drop") %>%
  # Calculate each taxon's share of the total count for that year
  group_by(Year) %>%
  mutate(proportion = total / sum(total) * 100) %>%
  ungroup() %>%
  # Keep only the study period of interest
  filter(Year >= 2020, Year <= 2025)

# Assign a distinct color to each taxon for the plot
colors <- c("Copepod" = "#E63946", "Ostracod" = "#2A9D8F",
            "Cladocera" = "#E9C46A", "Amphipod" = "#F4A261",
            "Chironomid" = "#457B9D")

# Pull out the final year's data to use for end-of-line labels
labels <- plot_data %>% filter(Year == max(Year))

# Build the plot
ggplot(plot_data, aes(x = Year, y = proportion, color = taxon)) +
  # Draw a line and point for each taxon across years
  geom_line(linewidth = 1.4) +
  geom_point(size = 2.5) +
  # Label each line directly at its endpoint instead of using a legend
  geom_text(data = labels, aes(label = taxon),
            hjust = -0.1, size = 3.2, fontface = "bold") +
  # Apply the custom color palette defined above
  scale_color_manual(values = colors) +
  # Show every year as a tick mark; expand right side to make room for labels
  scale_x_continuous(breaks = 2020:2025,
                     expand = expansion(mult = c(0.02, 0.25))) +
  # Format y-axis values as percentages
  scale_y_continuous(labels = scales::label_number(suffix = "%")) +
  # Add title (uses markdown bold via ggtext), subtitle, axis label, and caption
  labs(title = "**Aquatic invertebrate composition over time**",
       subtitle = "Proportion of five key taxa, 2020-2025",
       x = NULL, y = "% of total count",
       caption = "Data: Goleta Slough Aquatic Sampling") +
```

```r
# Start with a minimal theme, then customize for a dark background look
theme_minimal(base_size = 12) +
theme(
  # Dark navy background for both the plot panel and surrounding area
  plot.background  = element_rect(fill = "#0D1B2A", color = NA),
  panel.background = element_rect(fill = "#0D1B2A", color = NA),
  # Default all text to white so it's readable on the dark background
  text             = element_text(color = "white"),
  # element_markdown() needed here to render the ** bold ** in the title
  plot.title       = element_markdown(size = 14, color = "white"),
  plot.subtitle    = element_text(color = "#AAAAAA", size = 10),
  plot.caption     = element_text(color = "#666666", size = 8),
  axis.text        = element_text(color = "#AAAAAA"),
  # Subtle grid lines that show up on the dark background
  panel.grid.major = element_line(color = "#1E3A5F"),
  panel.grid.minor = element_blank(),
  # Hide the legend since lines are labeled directly
  legend.position  = "none"
)
```




```r
# Save the finished figure to the code folder as a high-resolution PNG
ggsave(here("code", "aquatic_viz.png"), width = 8, height = 5, dpi = 300)
```

4. Describing the process

I adapted Steven Ponce's ggplot2 timeseries code by replacing his energy dataset with aquatic invertebrate sampling data from Goleta Slough, adjusting the color palette, and modifying the axis labels and annotations to fit the new context. The biggest challenge was file path errors — getting R to find the data files correctly during rendering required switching from relative paths to the `here` package. I also had to find the exact column name for the Chironomid taxon since it had an unexpected trailing space. Unlike simpler ggplot figures from class, this visualization uses a custom dark theme, direct line labeling instead of a legend, and `ggtext` for markdown formatting in the title — techniques I had not used before.